

Deep Reinforcement Learning for Multi-Objective Resource Allocation in Multi-Platoon Cooperative Vehicular Networks

Yuanyuan Xu, *Member, IEEE*, Kun Zhu[✉], *Member, IEEE*, Hu Xu, and Jiequ Ji[✉]

Abstract—Grouping vehicles into platoons is a promising cooperative driving scenario to enhance the traffic safety and capacity of future vehicular networks. However, fast changing channel conditions in multi-platoon vehicular networks cause tremendous uncertainty for resource allocation. In addition, the unprecedented proliferation of various emerging vehicle-to-infrastructure (V2I) applications may result in some service demands with conflicting quality of experience. In this paper, we formulate a multi-objective resource allocation problem, which maximizes the transmission success ratio of intra-platoon communications and the mean opinion score (MOS) of V2I communication links. To efficiently solve this multi-objective optimization problem, we resort to a deep reinforcement learning (DRL) framework. Specifically, we divide it into a set of scalar optimization subproblems based on the weighted sum approach and model each one as a partially observable stochastic game (P-OSG), where each platoon acts as an agent and the actions taken by all platoons correspond to the resource allocation solution. We further propose a contribution-based dual-clip proximal policy optimization (CD-PPO) algorithm to deal with each subproblem, which is a DRL algorithm based on the actor-critic framework. The network parameters of all subproblems are then optimized collaboratively by using the proposed training algorithm and the neighborhood parameter transfer strategy. The desired Pareto front is obtained when all subproblems are solved. Simulation results reveal that the proposed algorithm can outperform other algorithms in terms of the MOS and transmission success ratio.

Index Terms—Multi-platoon vehicular networks, resource allocation, deep reinforcement learning, dual-clip proximal policy optimization.

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I. INTRODUCTION

THE blowout proliferation of vehicles has led to increasingly congested urban traffic, raising concerns about traffic safety and environmental pollution [1], [2]. Platooning, as one of the cooperative driving applications, is a promising technology for traffic management in future vehicular networks. Vehicles are grouped into platoons. Within each platoon, one vehicle acts as a leader while the other vehicles follow. With the aid of cellular-based vehicle-to-everything (C-V2X) communication technology, vehicles in the platoon can interact with each other. With shared driving information and information collected by local sensors, traffic conditions are predicted so that steering, accelerating, and braking of each follower can be automatically determined to improve the overall traffic safety and capacity and reduce fuel consumption [3], [4]. Furthermore, with such semi-autonomous driving, drivers can have more time to engage in various vehicle-to-infrastructure (V2I) applications, e.g., video streaming services for entertainment.

Despite the huge superiority of vehicular platooning, several technical problems should be carefully resolved before it becomes an effective traffic congestion solution [5], [6], [7]. In order to fully utilize the available resources, vehicle-to-vehicle (V2V) communication links always share spectrum with vehicle-to-infrastructure (V2I) connections [8]. Since vehicles in different platoons interfere with each other within their communication range over the same spectrum, the spectrum access decision of one platoon will affect that of other platoons. To improve communication quality, it is necessary to alleviate the interference between platoons. Besides, fast-changing channel conditions in high-dynamic vehicular networks bring considerable uncertainty for resource allocation [9]. If the instantaneous channel state information collected is inaccurate, resource allocation cannot be performed efficiently. Motivated by the aforementioned considerations, how to explore an effective resource allocation strategy in a multi-platoon vehicular network still remains to be resolved. In this paper, we investigate a joint subchannel selection and power allocation design in multi-platoon vehicular networks.

A. Related Work

Resource allocation problem in existing vehicular networks includes subchannel assignment and transmission power

control, which can be formulated as a joint optimization problem, e.g., simultaneously maximizing the throughput and reliability of V2X communications can be regarded as optimization objectives. The framework for solving such problems can be divided into two categories: (i) centralized framework and (ii) decentralized framework. The centralized framework can achieve good performance. Sun et al. in [10] considered the resource allocation problem for multicarrier non-orthogonal multiple access (MC-NOMA) systems, where the maximization of weighted throughput was formulated as a non-convex optimization problem. Using monotonic optimization, an optimal resource allocation policy can be obtained, and a sub-optimal solution with low computational complexity was developed as well based on successive convex approximation. The centralized framework needs to address the issue of communication overhead when collecting global information before extending it to vehicular network [11]. In recent years, decentralized and distributed frameworks have attracted significant attention for the resource allocation problem [12], [13]. Such optimization problems are generally NP-hard, and traditional optimization methods simply decompose them into multiple steps to achieve local optimal or suboptimal solutions, e.g., a Lyapunov-based optimization method in [14] and a novel graph-based algorithm in [15].

Recently, reinforcement learning (RL) based methods have been used in many literatures to deal with the resource allocation problem in wireless networks [16], [17], [18] and vehicular networks [19], [20], [21], [22], [23]. RL relies on iterative learning and decision-making process and is able to provide a promising framework to handle these problems through interacting with a dynamic environment for getting the maximum reward. Reference [24] proposed a deep reinforcement learning (DRL) method to improve spectrum efficiency. Each V2V link or each vehicle acted as an agent to select the subchannel and power level. The spectrum allocation and power control of different V2V links can be modeled as a multi-agent reinforcement learning problem. Reference [25] further adopted a multi-agent RL method to learn a distributed resource allocation strategy for improving the performance of both V2V and V2I communication links.

For platoon cooperation, most existing works investigated the intra-platoon communications based on C-V2X technology. Due to the growing communication hops in a platoon scenario, Dedicated Short Range Communications (DSRC) technology cannot guarantee the requirements for communication delay in a multi-platoon scenario [26]. C-V2X can provide high capacity, wide coverage, and high penetration. References [27] and [28] both investigated the capability of C-V2X technology to guarantee the latency and reliability in platoon scenarios, where a simple device-to-device (D2D) communication strategy was designed for cooperative awareness messages (CAM) dissemination within a platoon. Reference [29] utilized a detailed communication model that consisted of evolved multimedia broadcast multicast service (eMBMS) and D2D multicast communications. A joint resource allocation and platoon control method in LTE-vehicle-to-vehicle networks was proposed in [30].

This work aimed at simultaneously maintaining the string stability of the platoon and increasing the reliability of V2V communications. Reference [31] proposed a two-step resource allocation strategy considering platoon formation to maximize the platoon size and minimize the power consumption.

B. Motivation and Contribution

Most works ignored the impact of platoon formation and collaboration between multiple platoons on the efficiency and reliability of vehicular networks. The intra-platoon communications include various communication modes and multiple communication links. For example, in the widely adopted predecessor-leader following control strategy, each platoon member (PM) communicates with the preceding PM in unicast mode, while the platoon leader (PL) communicates with PM in multicast or broadcast mode. Besides ensuring the intra-platoon communication quality, mitigating the interference between multiple platoons should also be carefully investigated. Therefore, we emphasize the cooperation not only between vehicles within each platoon but also between different platoons. In addition, diverse service requirements for 5G V2X enhancements in Release 15 [32] have been proposed, such as advanced driving and mobile high data rate entertainment in 5G cellular V2X networks. For example, the entertainment applications need high bandwidth V2I connection to the base station (BS) for content download, video streaming and web browsing. Meanwhile, advanced driving in platoon needs to circularly disseminate cooperative awareness messages in each platoon through V2V communications, with high reliability. Therefore, a multi-objective optimization problem is considered in this work that maximizes the transmission success ratio of the intra-platoon communications and the mean opinion score (MOS) of V2I communications by jointly optimizing the subchannel selection and power allocation in a given finite period. The MOS methodology links the technical objective parameters with the subjective user perceived quality. Different MOS models have been proposed for different types of applications such as content download, video streaming, and web browsing [33], [34]. We concentrate on video streaming delivery for V2I links.

We explore a DRL-based framework to address the formulated multi-objective problem. RL provides a powerful tool for high maneuverability and uncertainty in the vehicular networks. The dynamic characteristics of platoons (i.e., vehicles leaving or joining the platoon) and the rapid changes of channel conditions both impose formidable challenges on the solution of the optimization problem. RL can find the optimal policy through the trial-and-error interaction with the unknown environment, and the corresponding policy can be seen as a feasible solution for the traditional optimization problem. In order to guarantee the real-time response, resource allocation has high demands on the computation time and generalization ability of the solution. Therefore, the multi-objective evolutionary methods, which have long been considered as suitable algorithms to address multi-objective

problems, are not suitable for the resource allocation problem in multi-platoon vehicular networks.

In the DRL-based framework, we decompose the multi-objective optimization problem into a number of scalar optimization subproblems and utilize a neighborhood-based parameter-transfer strategy to accelerate training. We finally can obtain a set of trade-off solutions, termed Pareto optimal solutions. Furthermore, we develop a distributed multi-agent RL algorithm for each subproblem. Multi-agent reinforcement learning (MARL) [35], [36], [37] can effectively solve these problems in a distributed manner, particularly when platoons only have individual local information. In particular, we model the subproblem as a partially observable stochastic game (POSG). To address the credit assignment problem of multiple agents, the contribution of each agent to the total accumulate reward is evaluated to encourage lazy agents to explore. In addition, considering that in the large-scale training environment, actions with negative advantages can make the new policy deviate significantly from the old policy, a contribution-based dual-clip proximal policy optimization (CD-PPO) algorithm has been proposed to solve the POSG.

The main novelty and contributions of this paper are summarized as follows:

- We develop a framework of multi-platoon vehicular networks for mobile high data rate entertainment and advanced driving. Then we formulate a multi-objective optimization problem that maximizes the transmission success ratio of the intra-platoon communications and the MOS of V2I communications.
- Considering the uncertainty in the environment (e.g., high mobility of vehicles and dynamics of platoons), a DRL-based framework is explored to solve our multi-objective optimization problem. Specifically, we decompose it into a number of scalar optimization subproblems and model each one as a neural network.
- To tackle these challenging subproblems, we formulate each subproblem as a POSG and propose a CD-PPO algorithm to solve it.
- Simulation results show the following several conclusions. Firstly, the proposed CD-PPO-based algorithm outperforms the PPO-based algorithm and the MARL-based algorithm in terms of the cumulative reward. Secondly, introducing the contribution of each agent into the reward design can alleviate the lazy agent problem, and some results also show that the dual-clip method can improve the traditional PPO algorithm to further accelerate and stabilize the convergence. Thirdly, comparing with other benchmark schemes with random subchannel selection or fixed power allocation, the proposed joint design achieves higher transmission success ratio and MOS.

C. Paper Organization

The remainder of this paper is organized as follows. In Section II, the particulars of the system model are introduced and the formulation of the resource allocation problem is developed. Section III presents the detail of the DRL-based framework to solve the multi-objective problem and the

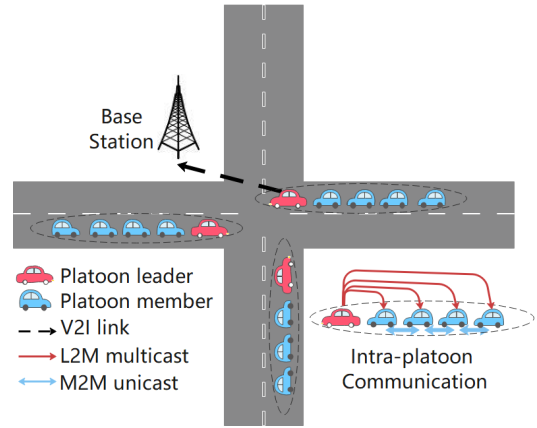


Fig. 1. System model of multi-platoon vehicular networks where multiple L2M and M2M links reuse the frequency spectrum preoccupied by V2I links.

CD-PPO algorithm to solve each subproblem. Simulation results and discussions are shown in Section IV. Finally, we conclude the paper in Section V.

II. SYSTEM DESCRIPTION

As shown in Fig. 1, we consider a multi-platoon vehicular communication network. Vehicles share the same path, and they are adaptively grouped into a platoon. For convenience, we apply a predecessor-leader following control strategy, where each platoon member (PM) communicates with other PMs and the preceding platoon leader (PL). Therefore, two modes exist in the intra-platoon communications: PL-to-PM (L2M) communications in multicast manner and PM-to-PM communications (M2M) in unicast manner. During time slot t , CAMs are propagated periodically in the platoon. On the basis of 3GPP cellular V2X architecture [32], V2I communication links provide simultaneous support for mobile high data rate entertainment and V2V communication links to ensure reliable driving information sharing for safety driving.

We assume that there are M subchannels and each of them is occupied by one of M V2I links. We define the set of V2I links as $\mathcal{M} = \{1, \dots, M\}$ and the set of platoons as $\mathcal{N} = \{1, \dots, N\}$, respectively. A platoon can be considered as a device-to-device (D2D) group. Let $\mathcal{D} = \{1, \dots, D\}$ denote the set of L2M receivers. Then the set of L2M receivers in the n -th platoon is denoted as $\mathcal{D}_n$. Similarly, we define the set of M2M links as $\mathcal{K}$ and the set of M2M links in the n -th platoon as $\mathcal{K}_n$, respectively. It is worth noting that the system employs an orthogonal frequency domain multiplexing (OFDM) protocol. We assume that M orthogonal spectrum subchannels have been allocated to M V2I links with fixed transmission power and the m -th V2I link occupies the m -th subchannel. In addition, L2M and M2M links reuse the frequency resources allocated to V2I links. Tab. I summarizes the notations and definitions used throughout this paper.

In this work, we aim to maximize the transmission success ratio of the intra-platoon communications and the MOS of V2I communications through subchannel selection and power allocation. Since the L2M, M2M, and V2I links transmitted on the same subchannel cause interference to each other,

TABLE I
NOTATIONS AND DEFINITIONS

Symbol	Definitions
P_{noise}	Noise power
ξ	Path-loss exponent
$\mathcal{N}$	Set of platoons
$\mathcal{M}$	Set of V2I links
$\mathcal{K}$	Set of M2M links
$\mathcal{D}$	Set of L2M receivers
$\mathcal{K}_n$	Set of M2M links in the n -th platoon
$\mathcal{D}_n$	Set of L2M receivers in the n -th platoon
$P_{n,m}^{\text{L2M}}$	Transmission power of L2M leader n
$P_{k,m}^{\text{M2M}}$	Transmission power of M2M transmitter k
P_m^{V2I}	Uplink transmission power of V2I transmitter m
$ H_{a,b} ^2$	Integrated channel power gain between two nodes
$G_n^{\text{L2B}}[m]$	Channel gain from L2M transmitter n to BS
$G_k^{\text{M2B}}[m]$	Channel gain from M2M transmitter k to BS
$G_{m,d}^{\text{V2I}}[m]$	Channel gain from V2I transmitter m to L2M receiver d
$G_{k,d}^{\text{M2M}}[m]$	Channel gain from M2M transmitter k to L2M receiver d
$G_{k,m}^{\text{V2I}}[m]$	Channel gain from V2I transmitter m to M2M receiver k
$G_{k,n}^{\text{L2M}}[m]$	Channel gain from L2M transmitter n to M2M receiver k

interference for different kinds of links is evaluated first for a given spectrum allocation and power allocation decision. Then the transmission rates for different links are expressed with respect to interference and transmission power. The transmission success ratio of the intra-platoon communications and MOS of V2I communications are further defined as functions of transmission rates.

A. Propagation and Interference Model

The propagation channels experience path-loss, Rayleigh fading, and log-normal shadowing. For example, the channel gain between nodes a and b , i.e., $G_{a,b}$, is described as follows:

$$G_{a,b} = |h_{a,b}|^2 d_{a,b}^{-\xi}, \quad (1)$$

where $|h_{a,b}|^2$ denotes the integrated channel power gain between nodes a and b due to the Rayleigh fading and log-normal shadowing, ξ is the path-loss exponent and $d_{a,b}$ is the distance between a and b .

Let $P_{n,m}^{\text{L2M}}$ and $P_{k,m}^{\text{M2M}}$ denote the transmission power of L2M leader n and M2M transmitter k over the m -th subchannel, respectively. In addition, the uplink transmission power of V2I transmitter m is denoted as P_m^{V2I} . They are subject to the following power constraints.

$$\begin{aligned} 0 &\leq P_{n,m}^{\text{L2M}} \leq P_{\max}^{\text{L2M}}, \\ 0 &\leq P_{k,m}^{\text{M2M}} \leq P_{\max}^{\text{M2M}}, \end{aligned} \quad (2)$$

where $P_{\max}^{\text{L2M}}$ and $P_{\max}^{\text{M2M}}$ are the maximum transmission power for L2M and M2M links, respectively.

The spectrum allocation is modeled as $\{a_n[m], b_k[m]\}$, with $a_n[m] = 1$ (or $b_k[m] = 1$) indicating that the n -th L2M leader (or the k -th M2M link) reuses the m -th subchannel, otherwise $a_n[m] = 0$ (or $b_k[m] = 0$). The interference model of multi-platoon vehicular networks can be elaborated as follows.

1) *Interference Caused to V2I*: Interference received at the m -th V2I link consists of the interference from L2M and M2M links that reuse the m -th subchannel:

$$I_m[m] = \sum_{n \in \mathcal{N}} a_n[m] P_{n,m}^{\text{L2M}} G_n^{\text{L2B}}[m] + \sum_{k \in \mathcal{K}} b_k[m] P_{k,m}^{\text{M2M}} G_k^{\text{M2B}}[m], \quad (3)$$

where $G_n^{\text{L2B}}[m]$ and $G_k^{\text{M2B}}[m]$ denote the channel gains from the n -th L2M transmitter to the BS and from the k -th M2M transmitter to the BS over the m -th subchannel, respectively.

2) *Interference Caused to L2M*: Interference received at receiver d in the n -th L2M multicast link over the m -th subchannel is given by:

$$I_{n,d}[m] = P_m^{\text{V2I}} G_{m,d}^{\text{V2I}}[m] + \sum_{k \in \mathcal{K}} b_k[m] P_{k,m}^{\text{M2M}} G_{k,d}^{\text{M2M}}[m] + \sum_{n' \neq n} a_{n'}[m] P_{n',m}^{\text{L2M}} G_{n',d}^{\text{L2M}}[m], \quad (4)$$

where $G_{m,d}^{\text{V2I}}[m]$ and $G_{k,d}^{\text{M2M}}[m]$ denote the channel gains from V2I transmitter m and M2M transmitter k to L2M receiver d over the m -th subchannel, respectively. $G_{n',d}^{\text{L2M}}[m]$ denotes the channel gain from L2M transmitter n' in the n' -th platoon to receiver d over the m -th subchannel.

3) *Interference Caused to M2M*: The interference of the k -th M2M link over the m -th subchannel is given by:

$$I_k[m] = P_m^{\text{V2I}} G_{k,m}^{\text{V2I}}[m] + \sum_{n \in \mathcal{N}} a_n[m] P_{n,m}^{\text{L2M}} G_{k,n}^{\text{L2M}}[m] + \sum_{k' \neq k} b_{k'}[m] P_{k',m}^{\text{M2M}} G_{k,k'}^{\text{M2M}}[m], \quad (5)$$

where $G_{k,m}^{\text{V2I}}[m]$ and $G_{k,n}^{\text{L2M}}[m]$ denote the channel gains from the m -th V2I link and the n -th L2M transmitter to M2M receiver k over the m -th subchannel, respectively. $G_{k,k'}^{\text{M2M}}[m]$ denotes the channel gain from the k' -th M2M transmitter to M2M receiver k over the m -th subchannel.

B. Transmission Model

1) *V2I Link*: The uplink transmission rate of the m -th V2I link can be expressed as

$$R_m^{\text{V2I}}[m] = \log_2 \left(1 + \frac{P_m^{\text{V2I}} G_m^{\text{BS}}[m]}{P_{\text{noise}} + I_m[m]} \right), \quad (6)$$

where P_{noise} denotes the noise power and $G_m^{\text{BS}}[m]$ denotes the channel gain from the m -th V2I transmitter to the BS over the m -th subchannel.

2) *L2M Link*: The channel quality of the receiver d in the n -th L2M multicast link over the m -th subchannel is given by

$$\beta_{n,d}^{\text{L2M}}[m] = \frac{G_{n,d}^{\text{L2M}}[m]}{P_{\text{noise}} + I_{n,d}[m]}, \quad (7)$$

where $G_{n,d}^{\text{L2M}}[m]$ denotes the channel gain of L2M receiver d from the L2M transmitter n over the m -th subchannel. For the n -th L2M multicast link, its transmission condition in the

m -th subchannel is determined by the receiver with the worst condition [38]:

$$\beta_n^{\text{L2M}}[m] = \min_{d \in D_n} \beta_{n,d}^{\text{L2M}}[m]. \quad (8)$$

The normalized transmission rate of the n -th L2M multicast link is given by

$$r_n^{\text{L2M}}[m] = \log_2(1 + P_{n,m}^{\text{L2M}} \beta_n^{\text{L2M}}[m]). \quad (9)$$

Thus the aggregate transmission rate of the n -th L2M multicast link is given by

$$R_n^{\text{L2M}}[m] = |D_n| r_n^{\text{L2M}}[m]. \quad (10)$$

3) *M2M Link*: The transmission rate of the k -th M2M link over the m -th subchannel can be expressed as

$$R_k^{\text{M2M}}[m] = \log_2(1 + \frac{P_{k,m}^{\text{M2M}} G_{k,k}^{\text{M2M}}[m]}{P_{\text{noise}} + I_k[m]}), \quad (11)$$

where $G_{k,k}^{\text{M2M}}[m]$ denotes the channel gain of the k -th M2M receiver from the k -th M2M transmitter over the m -th subchannel.

C. Problem Formulation

We divide the constant duration T into a series of discrete time slots with equal length. Accordingly, the subchannel selection and power control of intra-platoon communications in the n -th platoon are denoted as

$$\begin{aligned} \mathbf{A}_n &= \{(a_n[m]^{(1)}, P_{n,m}^{\text{L2M}(1)}), \dots, (a_n[m]^{(t)}, P_{n,m}^{\text{L2M}(t)}), \\ &\quad \dots, (a_n[m]^{(T)}, P_{n,m}^{\text{L2M}(T)})\}, \\ \mathbf{B}_n &= \{(b_k[m]^{(1)}, P_{k,m}^{\text{M2M}(1)}), \dots, (b_k[m]^{(t)}, P_{k,m}^{\text{M2M}(t)}), \\ &\quad \dots, (b_k[m]^{(T)}, P_{k,m}^{\text{M2M}(T)})\}, k \in \mathcal{K}. \end{aligned} \quad (12)$$

The index t in $a_n[m]^{(t)}$ or $P_{n,m}^{\text{L2M}(t)}$ represents the current time slot of decision, and $T = |T|$ is the scheduling horizon ahead for decision making. $\mathbf{A}_n$ and $\mathbf{B}_n$ represent the subchannel selection and power control of the L2M and M2M links in the n -th platoon, respectively.

V2V links are used to deliver CAM periodically among neighboring vehicles. Thus, a function $\mathcal{S}$ is defined to mark whether a transmission can be finished within a time slot t :

$$\mathcal{S}(R_n^{\text{L2M}}[m]^{(t)}) = \begin{cases} 1, & \text{if } R_n^{\text{L2M}}[m]^{(t)} * t \geq B, \\ 0, & \text{otherwise,} \end{cases} \quad (13)$$

where B represents the size of the CAM data. Then, the problem of maximizing the transmission success ratio of intro-platoon communications in each platoon can be mathematically described as

$$\max_{\mathbf{A}, \mathbf{B}} f_1 = \sum_{n \in \mathcal{N}} \frac{\sum_{m \in \mathcal{M}} \mathcal{S}(R_n^{\text{L2M}}[m]^{(t)}) + \sum_{k \in \mathcal{K}} \mathcal{S}(R_k^{\text{M2M}}[m]^{(t)})}{|\mathcal{N}| + |\mathcal{K}|}, \quad (14)$$

where $\mathbf{A} = [\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_n, \dots, \mathbf{A}_N]$ and $\mathbf{B} = [\mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_n, \dots, \mathbf{B}_N]$ are matrices of decision vectors $\mathbf{A}_n$ and $\mathbf{B}_n$ for $n \in \mathcal{N}$, respectively.

Moreover, V2I communication links are responsible for providing entertainment services with high mobile transmission rate. Inspired by the widely used QoE metric, the mean opinion score (MOS) model is used to measure users' QoE for services such as content download, video streaming, and web browsing. We concentrate on video streaming delivery for V2I links. The transmission delay is an important performance indicator of mobile network, and the delay of m -th V2I link is calculated by dividing the amount of data by the transmission rate. The value of MOS depends on the transmission delay, and the MOS of m -th V2I link is denoted as follows [39].

$$\text{MOS}_m = C_1 \ln\left(\frac{1}{B_m/R_m^{\text{V2I}}[m]}\right) + C_2, \quad (15)$$

where C_1 and C_2 are both constants and $C_1 > 0$. B_m is the transmission data size of the m -th V2I link. Smaller delay results in a higher MOS and better users' QoE.

As such, the long-term MOS of the whole vehicular network is defined as follows.

$$\max_{\mathbf{A}, \mathbf{B}} f_2 = \sum_{t \in T} \sum_{m \in \mathcal{M}} \text{MOS}_m^{(t)}. \quad (16)$$

Accordingly, the resource allocation problem aims to simultaneously maximize the transmission success ratio of intra-platoon communications and the MOS of V2I communications by jointly designing subchannel selection and power allocation. Therefore, the multi-objective problem can be mathematically formulated as

$$\max_{\mathbf{A}, \mathbf{B}} f = [f_1, f_2] \quad (17a)$$

$$\text{s.t. } a_n[m]^{(t)} = \{0, 1\}, \forall n \in \mathcal{N}, m \in \mathcal{M}, \quad (17b)$$

$$b_k[m]^{(t)} = \{0, 1\}, \forall k \in \mathcal{K}, m \in \mathcal{M}, \quad (17c)$$

$$\sum_{m \in \mathcal{M}} a_n[m]^{(t)} \leq 1, \forall n \in \mathcal{N}, \quad (17d)$$

$$\sum_{m \in \mathcal{M}} b_k[m]^{(t)} \leq 1, \forall k \in \mathcal{K}, \quad (17e)$$

$$0 \leq P_{n,m}^{\text{L2M}(t)} \leq P_{\max}^{\text{L2M}}, \forall n \in \mathcal{N}, m \in \mathcal{M}, \quad (17f)$$

$$0 \leq P_{k,m}^{\text{M2M}(t)} \leq P_{\max}^{\text{M2M}}, \forall k \in \mathcal{K}, m \in \mathcal{M}. \quad (17g)$$

(17b)-(17e) are the subchannel allocation constraints that each L2M or M2M link only reuses at most one subchannel, where $a_n[m]^{(t)}$ and $b_k[m]^{(t)}$ are binary variables indicating whether the n -th L2M leader and the k -th M2M link reuse the m -th subchannel at time slot t , respectively. (17f) and (17g) are the transmission power constraints of L2M and M2M links, where $P_{n,m}^{\text{L2M}(t)}$ and $P_{k,m}^{\text{M2M}(t)}$ represent the transmission power of the n -th L2M leader and the k -th M2M link using the m -th subchannel at time slot t , respectively. There exist two challenges in solving this problem. The first difficulty lies in the multiple optimization objectives that are conflicting. The second one is that it is impractical to solve with traditional optimization techniques due to the uncertainty and high dynamics in the multi-platoon vehicular environment.

The work in [25] utilized a fingerprint-based deep Q network (DQN) based MARL algorithm to solve the resource allocation problem in vehicular environments with high mobility, where a link-level agent was used to determine discrete

power control and subcarrier selections. If changes in the number of links due to vehicle dynamics are not considered in advance, the neural network trained previously is only suitable for a fixed number of links. This is because that if the input and output dimensions of this agent change, the neural network needs to be retrained. In this paper, the proposed resource allocation method considers the dynamics of platoons. Through restricting the input and output dimensions of the neural network to remain constant by fixing the maximum size of a platoon, we can apply the algorithm to the scenarios with a large number of vehicles and vehicles joining and leaving platoons dynamically.

III. PROPOSED RESOURCE ALLOCATION METHOD

The problem in (17) is an NP problem due to the following reasons. Firstly, (17b) and (17c) are binary constraints, which introduce non-convex and combinatorial characteristics into the formulated multi-objective optimization problem. Secondly, the subchannel selection constraints (17d) and (17e) are non-convex with respect to scheduling variables $a_n[m]$ and $b_k[m]$. There are several classical algorithms (e.g., Pareto local search algorithm and multiple objective genetic local search algorithm) that can be used to solve this problem, but they have high computational complexity, especially for large-scale scenarios. Monotonic optimization and successive convex approximation in [10] can be used to develop a centralized solution with low complexity. However, the time-varying impact of the wireless channels on the optimal decision process in dynamic environments poses an intractable challenge to traditional optimization algorithms, as these traditional solutions often require re-running the algorithms once the environment changes. Thus, we propose a DRL-based framework to solve this multi-objective optimization problem, where it is decomposed into a series of scalar optimization subproblems by applying the weighted sum approach. We then model each subproblem as a POSG and resort to a CD-PPO-based algorithm to solve it. In addition, we optimize the network parameters of all subproblems collaboratively using the neighborhood-based parameter transfer method and the proposed training algorithm.

A. DRL-Based Framework for Multi-Objective Optimization

The decomposition strategy is used as the basic framework, we first decompose the multi-objective optimization problem into a number of scalar optimization subproblems. We use the weighted sum method, where a set of uniform weight vectors $\lambda^1, \dots, \lambda^N$ is adopted, e.g., $(1, 0), (0.99, 0.01), \dots, (0, 1)$ for a bi-objective problem. Hence, we have $\lambda^n = (\lambda_1^n, \dots, \lambda_L^n)$, where L is the number of objectives. Then the original multi-objective problem is divided into N scalar subproblems. We model each subproblem as a POSG and use the reinforcement learning algorithm to solve it, where training time of subproblems are reduced using the neighborhood-based parameter transfer strategy. When all subproblems are solved, the Pareto optimal solution and Pareto Front can be obtained [40].

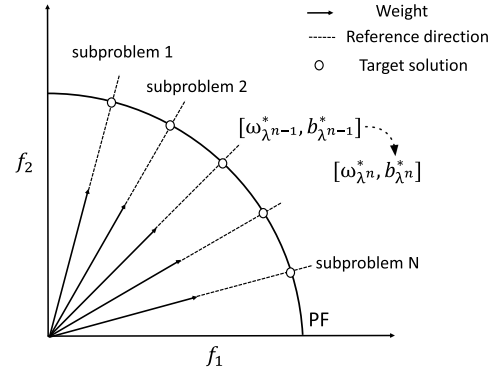


Fig. 2. Illustration of the decomposition and parameter-transfer strategy.

In particular, the objective of the n^{th} subproblem is

$$\max \lambda^n \cdot f = \sum_{l=1}^L \lambda_l^n \cdot f_l. \quad (18)$$

To efficiently solve these subproblems, we model each one as a neural network. The N scalar subproblems are then solved collaboratively via the proposed actor-critic algorithm and the neighborhood-based parameter transfer strategy. According to (18), we observe that any two neighboring subproblems may have very close solutions, because they have adjacent weight vectors. Each subproblem can draw the support from the knowledge of its neighboring subproblem to solve itself. Under such a DRL framework, the network parameters of the $(n-1)^{th}$ subproblem can be expressed as $[\omega_{\lambda^{n-1}}^*, b_{\lambda^{n-1}}^*]$, where $[\omega^*, b^*]$ represents the optimal parameters of the subproblem and $[\omega, b]$ represents the parameters that haven't been optimized. In addition, the network parameters of the $(n-1)^{th}$ subproblem achieved by the proposed algorithm are assumed to be asymptotically optimal. The optimal network parameters $[\omega_{\lambda^{n-1}}^*, b_{\lambda^{n-1}}^*]$ of the $(n-1)^{th}$ subproblem obtained are defined as the starting training point for the next n^{th} subproblem. As expressed, the network parameters will transfer from the previous subproblem to the next one in proper sequence, saving considerable amount of time compared to training all subproblems separately, as illustrated in Fig. 2.

The DRL framework for solving the multi-objective optimization problem is summarized in Algorithm 1, which contains the multi-objective decomposition and the neighborhood-based parameter transfer strategy. In this system, each subproblem is modeled as a POSG and solved by our proposed CD-PPO algorithm, and all the subproblems can be solved sequentially by transferring the network weights. Pareto optimal solution can be obtained by efficiently solving each subproblem. Moreover, when all subproblems are solved in sequence, we can obtain the desired Pareto Front. The DRL-based framework has two main advantages. 1) It is simple and modularized to use. Any DRL based solvers (including the adopted PPO algorithm) can be integrated to handle the multi-objective problems. 2) Once the trained model is available, we can obtain the approximate Pareto Front directly through a simple forward propagation of the model.

The DRL-based framework acts as an outer-loop. The next issue is how to model and solve those decomposed scalar

Algorithm 1 DRL Framework for Solving Multi-Objective Optimization

```

1: Input: The model of the subproblem  $M = [\omega, b]$ , weight
   vectors  $\lambda^1, \dots, \lambda^N$ 
2: Process:
3: Random initialize  $[\omega_{\lambda^1}, b_{\lambda^1}]$ 
4: for  $n = 1, \dots, N$  do
5:   if  $i == 1$  then
6:     Solve the first subproblem using CD-PPO algorithm
       and obtain the network parameters  $[\omega_{\lambda^1}^*, b_{\lambda^1}^*]$ 
7:   else
8:      $[\omega_{\lambda^n}, b_{\lambda^n}] \leftarrow [\omega_{\lambda^{n-1}}^*, b_{\lambda^{n-1}}^*]$ 
9:     Solve the  $n^{th}$  subproblem using CD-PPO algorithm
       and obtain the network parameters  $[\omega_{\lambda^n}^*, b_{\lambda^n}^*]$ 
10:  end for
11: Output: The approximate Pareto Front can be directly
    calculated by  $[\omega^*, b^*]$ 

```

optimization subproblems. In the next subsection, we model each subproblem as a POSG and propose a CD-PPO-based algorithm to solve it.

B. Preliminaries of POSG

In each subproblem learning system, we specify each platoon as the policy maker, which controls an agent to learn and update experiences from the environment. Each platoon can choose the spectrum and control the power for all intra-platoon communication links. The resource allocation in the multi-platoon environment can be modeled as a POSG, which can be described by the following tuple $(\mathcal{N}, \mathcal{S}, \{\mathcal{A}^n\}_{n=1}^N, \{\mathcal{O}^n\}_{n=1}^N, \mathcal{P}, \{\mathcal{R}^n\}_{n=1}^N)$, where $\mathcal{N}$ and $\mathcal{S}$ are environment agent set and state set, respectively. $\mathcal{A} = \mathcal{A}^1 \times \mathcal{A}^2 \times \dots \times \mathcal{A}^N$ represents the action space. $\mathcal{O} = \mathcal{O}^1 \times \mathcal{O}^2 \times \dots \times \mathcal{O}^N$ represents the observation space set. $\mathcal{P}(s'|s, a)$ is the function of transition probability. $\mathcal{R}^n : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathcal{R}$ represents the reward function of the n^{th} agent. When agents share the same reward function, i.e., $\mathcal{R}^1 = \mathcal{R}^2 = \dots = \mathcal{R}^N$, the problem becomes fully cooperative.

Independent Q-learning [41], a MARL-based algorithm, is unable to handle the non-stationary environment. Additionally, if we consider a cooperative MARL-based problem and set a global reward for all agents, some agents cannot quantify their contribution to the reward, causing the credit assignment problem. In other words, some lazy agents will lose the ability to explore, because other agents have learned good strategies to guarantee the system performance. Since other agents have done so well, the exploration of the agent may reduce the performance, and thus the exploration of this lazy agent becomes daunting. Therefore, it is necessary to quantify the contribution of each platoon to the system performance via the transmission success ratio.

C. Details of POSG

In our modeled stochastic game model, several key elements are described below in detail.

1) *Agent:* We specify the platoon leader, the controller in each platoon, as an agent, and the multi-platoon vehicular network defined in Section II as the learning environment. Each platoon acts as a policy maker to perform subchannel allocation and power control.

2) *State and Observation Space:* At each time slot, each agent observes the state information of the environment to make the corresponding policy. The state S_t at time slot t includes global channel conditions and all agents' behaviors, which is unknown to each platoon agent. For the state space, calculating the transmission rate for each communication link involves channel gain and interference, which can only be observed locally and is not known by other agents. Each agent partially observes local channel information, and the observation space of n^{th} agent is

$$\mathcal{O}^n = \{G_n, B_n, t\}, \quad (19)$$

where $G_n = \{G_{n,d}^{L2M}, G_{m,d}^{V2I}, G_k^{M2M}, G_{k,n}^{L2M}, G_{k,d}^{M2M}, G_{n,m}^{L2B}, G_{k,m}^{M2B}\}$, and $B_n = \{B_{L2M_n}, B_{M2M_1}, \dots, B_{M2M_k}\}, \forall k \in \mathcal{K}_n, \forall m \in \mathcal{M}, \forall d \in \mathcal{D}_n$. The intra-platoon communication consists of one L2M communication and several M2M communications. The observation space $\mathcal{O}^n$ is composed of signal channels, interference channels from other L2M or M2M transmitters, all V2I transmitters and from own transmitter to the BS. $G_{n,m}^{L2B}$ and $G_{k,m}^{M2B}$ are estimated at the BS, and transmitted to PL with small signaling overhead. Note that the remaining channel condition can be estimated by all PMs as the receiver of L2M or M2M links at the beginning of each time slot t , and then transmitted to the PL via delay-free feedback. These observations need to be normalized by subtracting the mean and dividing by the standard deviation using the statistics aggregated over the course of the entire training. Through normalization, we facilitate the elimination of the effect of large differences between variables on model performance. B_n represents the remaining percentage of CAM data of L2M and all M2M links in intra-platoon communication, and t represents the percentage of the time remaining over the transmission latency.

3) *Action Space:* $\mathcal{A}^n = \{A_{L2M_n}, A_{M2M_1}, \dots, A_{M2M_k}\}, \forall k \in \mathcal{K}_n$. A_{L2M} or A_{M2M_k} is a two-tuple: $\{S, P\}$. S is the subchannel that the L2M or M2M link selects, represented by an integer in the range of $[1, M]$. P denotes the transmission power of L2M or M2M communications, which is a continuous value and satisfies the constraints (17f) and (17g). When the power equals to 0, we set the transmission power to -100dBm.

It is worth noticing that the action definition contains discrete and continuous variables, which are not directly accessible to the learning algorithm. To solve this problem, we convert the discrete variable S to a continuous one. $S \in [0, 1]$ is the channel state indicator. If a continuous variable S is used, the platoon will choose $\lceil S * M \rceil$ -th subchannel for communication with fusion center, where $\lceil \cdot \rceil$ is the ceiling function.

4) *Reward Design:* In addition to the state transition, the environment can feed back an immediate reward. Reward design is vital for resolving problems with difficult-to-optimize objectives, and the system performance is critically interrelated

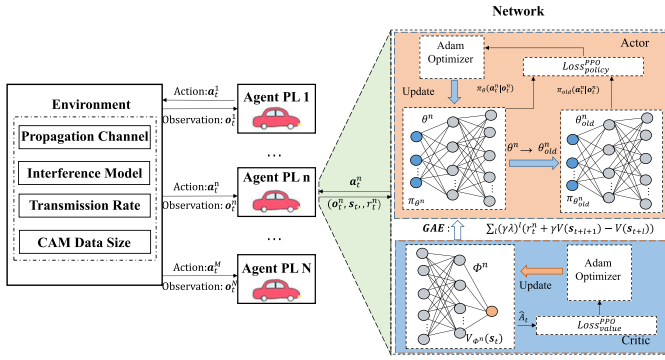


Fig. 3. Illustration of distributed MADRL framework for multi-platoon networks.

with rewards. In (17a), the proposed bi-objective problem includes maximizing the transmission success ratio of all intra-platoon communications in platoons and maximizing the MOS of all V2I links. In response to the first objective, we set the rewards equal to the L2M and M2M transmission rates until the CAM is delivered, after which the rewards of L2M and M2M are set to constants α and β that are larger than the largest possible L2M and M2M transmission rates, respectively. We encourage each L2M and M2M transmission to be finished as soon as possible, so the reward design is not only simply about transmission rate but also attempts to achieve more rewards of α and β . As such, the L2M-related and M2M-related rewards at each time step are set as

$$\mathcal{R}_{L2M_n}^{(t)} = \begin{cases} R_n^{L2M}[m]^{(t)}, & \text{if } B_{L2M_n} > 0, \\ \alpha, & \text{otherwise,} \end{cases} \quad (20)$$

and

$$\mathcal{R}_{M2M_k}^{(t)} = \begin{cases} R_k^{M2M}[m]^{(t)}, & \text{if } B_{M2M_k} > 0, \\ \beta, & \text{otherwise,} \end{cases} \quad (21)$$

respectively.

For the second objective, we naturally consider the MOS of all V2I links, and the global reward $\mathcal{R}_{t+1}$ is denoted as

$$\begin{aligned} \mathcal{R}_{t+1} = & \lambda_1 \left(\sum_{n \in \mathcal{N}} \mathcal{R}_{L2M_n}^{(t)} + \sum_{k \in \mathcal{K}} \mathcal{R}_{M2M_k}^{(t)} \right) \\ & + \lambda_2 \sum_{m \in \mathcal{M}} \text{MOS}_m^{(t)}, \end{aligned} \quad (22)$$

where λ_1 and λ_2 are positive weights for balancing two objectives. We utilize the transmission success ratio of communications in each platoon and quantify each agent's contribution to the global reward. The reward of n^{th} agent is defined as

$$\mathcal{R}_{t+1}^{(n)} = \frac{\mathcal{R}_{L2M_n}^{(t)} + \sum_{k \in \mathcal{K}_n} \mathcal{R}_{M2M_k}^{(t)}}{\alpha + |\mathcal{K}_n| \beta} \mathcal{R}_{t+1}. \quad (23)$$

D. Distributed Multi-Agent DRL Approach

Considering that the multi-platoon vehicular network is dynamic and complex such that the state space and action space become considerably large, we investigate a multi-agent DRL with local states and use the deep neural network (DNN)

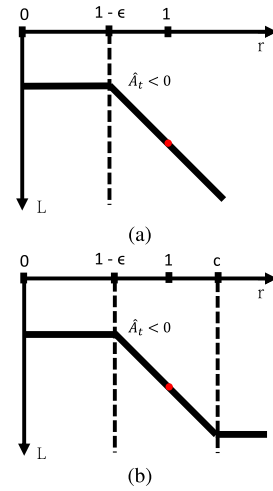


Fig. 4. (a) Standard PPO (clip with ϵ); (b) Dual-clip PPO (clip with ϵ and c where $\hat{A}_t < 0$).

as a function approximator, to tackle the challenges of large spaces. More specifically, a centralized-training-decentralized-execution paradigm is adopted in the MADRL framework.

The whole schematic of the distributed MADRL framework for the multi-platoon network is illustrated in Fig. 3. At the beginning of each episode, each platoon starts with a randomly initialized environment state and a CAM of data size B for intra-platoon communication. The length of episode equals to the delivery latency constraint T . The change of small-scale fading triggers the transition of the state and causes each platoon agent to select its actions. At each time slot, each platoon receives its own observation $\mathcal{O}_t^{(n)}$, i.e., the channel information, remaining CAM data size and the remaining time over transmission latency. Then agent n chooses the action generated from its policy π_{θ} . For this, agent n is employed with an actor-critic based DNN that takes the observation $\mathcal{O}_t^{(n)}$ as input and outputs the probability distribution of the policy. The actor DNN generates action sequences from a Gaussian distribution from given states and updates based on a policy gradient method. As an auxiliary module, the critic DNN uses a value-based method to generate temporal difference errors, which can be used to guide the gradient of the actor network to move toward the direction with a low cost.

The proposed MADRL based resource allocation approach is summarized in Algorithm 2. In the distributed execution phase, each agent first selects an action according to the probability density function, and all agents generate a joint action $\mathcal{A}_t$ to interact with the environment and receive a global reward $\mathcal{R}_{t+1}$. Then, each platoon agent gets $\mathcal{R}_{t+1}^{(n)}$ depending on the transmission success ratio of intra-platoon communications according to (23). Finally, each agent stores the experiences in the replay memory pool in parallel, and then the proposed CD-PPO algorithm is used to train the actor-critic network.

E. Training With Improved CD-PPO

Reinforcement learning aims to find a strategy to obtain the maximum return. The policy gradient algorithms optimize

Algorithm 2 MADRL Based Resource Allocation in Multi-Platoon Networks

```

1: Initialize value function parameters  $\phi$  and policy function
   parameters  $\pi$  randomly
2: for each iteration do
3:   while Platoons are within the range of BS do
4:     Update the position of vehicles and the large-scale
       fading of channel
5:     Reset CAMs  $B_n$ , for all  $n \in N$ 
6:     for time slot  $t = 1, \dots, T$  do
7:       for each platoon agent  $n$  do
8:         Observe  $\mathcal{O}_t^{(n)}$  and select action  $\mathcal{A}_t^{(n)}$  through
           importance sampling density function
9:       end for
10:      All agents execute a joint action  $\mathcal{A}_t$  and interact
        with the environment for receiving their reward
         $\mathcal{R}_{t+1}$ 
11:      Update channel small-scale fading
12:      for each platoon agent  $n$  do
13:        Compute  $\mathcal{R}_{t+1}^{(n)}$  according to (23), calculate the
          advantages  $\hat{A}_i^t$  using GAE method
14:      end for
15:    end for
16:  end while
17:  for each platoon agent  $n$  do
18:    Update policy function  $\theta$  by Adam w.r.t  $L_{\text{policy}}^{\text{PPO}}(\pi)$ 
19:    Update value function  $\phi$  by Adam w.r.t  $L_{\text{value}}^{\text{PPO}}(\phi)$ 
20:  end for
21: end for

```

the policy directly. The policy is modeled with a parameterized function with respect to π_θ . A method called trust region policy optimization (TRPO) [42] can guarantee that the current policy does not diverge sharply from the old one, via restricting the policy update by a Kullback-Leibler (KL) divergence. However, the training of TRPO is time-consuming due to its insufficient second-order optimization. Proximal Policy Optimization (PPO) [43] simplifies it by using a clipped surrogate objective while retaining similar performance.

1) *Standard PPO*: We use the sampled trajectories to construct the loss of value and policy function: $L_{\text{value}}^{\text{PPO}}(\theta)$ and $L_{\text{policy}}^{\text{PPO}}(\pi)$. The value function is used to estimate the advantage $\hat{A}_i^t$ using generalized advantage estimation (GAE) [44]. We construct the squared-error loss $L_{\text{value}}^{\text{PPO}}(\theta)$, and optimize it with the Adam Optimizer. We first express the probability ratio between old and new policies with

$$r(\theta) = \frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}. \quad (24)$$

Because the ratio $r(\theta)$ can be extremely large, maximization of the objective may lead to an excessively large policy deviation. To address this issue, the standard PPO imposes the constraint by forcing $r(\theta)$ to stay within a small interval around 1, precisely $[1 - \epsilon, 1 + \epsilon]$, where ϵ is a hyperparameter, to penalize

extreme changes to the policy. It is expressed as follows.

$$L_{\text{policy}}^{\text{PPO}}(\pi) = E[\min(r(\theta)\hat{A}_{\theta_{\text{old}}}(s, a), \text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_{\theta_{\text{old}}}(s, a))], \quad (25)$$

where the function $\text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)$ clips the ratio to be no more than $1 + \epsilon$ and no less than $1 - \epsilon$.

2) *Dual-Clip PPO*: However, in our large-scale training environment, we found that using actions with negative advantages can have an adverse effect. For example, when $\pi_\theta(a|s) \gg \pi_{\theta_{\text{old}}}(a|s)$, the ratio $r(\theta)$ is a large number. When $\hat{A} < 0$, such a large ratio will introduce a big and unbounded variance due to $r(\theta)\hat{A} \ll 0$. It's difficult to guarantee the policy convergence. So we further employ a dual-clipped function to support large-scale distributed learning, which clips the ratio $r(\theta)$ with a lower bound of the value of $r(\theta)\hat{A}$, as depicted in Fig. 4. When $\hat{A} < 0$, the new objective of dual-clipped PPO [45] is

$$L_{\text{policy}}^{\text{CD-PPO}}(\pi) = E[\max(\min(r(\theta)\hat{A}_{\theta_{\text{old}}}(s, a), \text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_{\theta_{\text{old}}}(s, a)), c\hat{A}_{\theta_{\text{old}}}(s, a))], \quad (26)$$

where $c > 1$ is a constant indicating the lower bound.

IV. SIMULATION RESULTS

In this section, we first validate the effectiveness of the MADRL based resource allocation approach in the considered network. Then the advantage of the CD-PPO algorithm is verified compared to existing methods. Finally, we investigate the performance of DRL framework for the multi-objective optimization problem.

A. Simulation Setup

We consider a multi-platoon vehicular system with $M = 25$ vehicles that can arbitrary leave their original platoon or join another new platoon. The parameter setting for V2I and V2V communications follows the existing 3GPP specification [46], which is shown in Table II.

B. Network Architecture

We conduct the experimental simulations using a workstation with a Geforce GTX 1050 GPU. The software platform of the experiment is Python 3.6 with PyTorch [47]. The proposed CD-PPO-based algorithm consists of the actor network and the critic network, each of which has one input layer, two hidden layers, and one output layer. Moreover, each hidden layer is assumed to have the same number of 1500 neurons. We use the rectified linear unit (ReLU) function $f_{\text{ReLU}}(x) = \max\{0, 1\}$ as the activation function in each hidden layer. The Adam optimizer is used to update the actor network and the critic network. The learning rate for both neural networks is set to 0.0001. The clip parameter and the discount factor in the proposed algorithm are set to $\epsilon = 0.2$ and $\gamma = 0.999$, respectively. In addition, the training process of our proposed algorithm has $N_{\text{ept}} = 20\,000$ episodes.

TABLE II
NUMERICAL CALCULATION PARAMETER SETTINGS

Parameter	Value	Parameter	Value
Maximum of Platoon Size	8	Number of Subchannels	25
Carrier Frequency	2GHz	Total Bandwidth	10MHz
V2I Transmit Power	23dBm	L2M Maximum Transmit Power	23dBm
M2M Maximum Transmit Power	23dBm	Noise Power	-114dBm
BS Antenna Height	25m	BS Antenna Gain	8dBi
BS Receiver Noise Figure	5dB	Latency constraint	10ms
Vehicle Antenna Height	1.5m	Vehicle Antenna Gain	3dBi
Vehicle Receiver Noise Figure	9dB	Lane Width	3.5m
Number of Lanes	4 in each direction	Number of Direction	4
Decorrelation Distance	10m	Path Loss	LOS NLOS
Shadowing distribution	Log-normal	Fast Fading	Rayleigh fading
Path Loss and Shadowing	Every 100ms	Fast Fading Update	Every 1ms

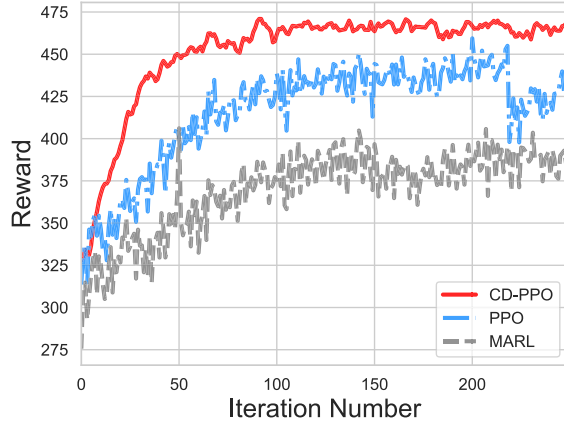


Fig. 5. Cumulative reward versus the number of training iterations for different algorithms.

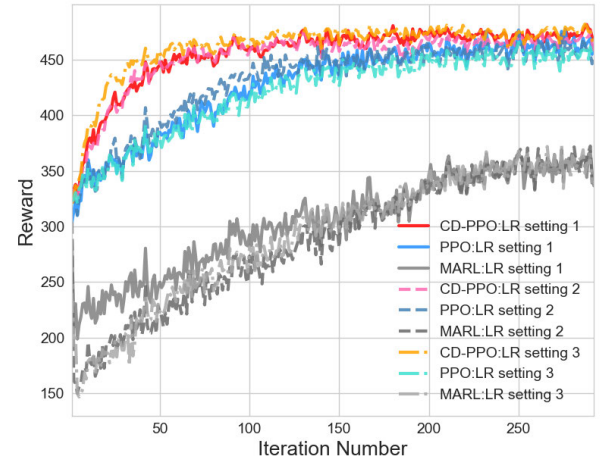


Fig. 6. Algorithm performance under learning rates of 0.0001, 0.001 and 0.00001 for settings 1, 2, and 3, respectively.

C. Result Analysis

For the sake of performance evaluation, we compare our proposed CD-PPO-based algorithm with the following algorithms.

- Standard PPO-based algorithm: Using a clipped surrogate objective function.
- MARL-based algorithm [25]: Using a fingerprint-based deep Q-learning method.

According to Algorithm 2, the computational complexity of the proposed CD-PPO, the PPO and the MARL algorithms in [25] is $\mathcal{O}(TN_l)$, where T represents the number of time slots and N_l is the number of all links. Fig. 5 shows comparison of three algorithms in terms of the attainable reward. We plot the same global reward performance. As seen from Fig. 5, the proposed algorithm achieves larger reward and converges faster compared to the benchmark algorithms, since the actor-critic network has been efficiently trained using the proposed method. Particularly, the training process using dual-clip is smooth while the standard PPO is choppy. Additionally, we observe that the reward curve of our proposed algorithm tends to smooth when the number of iterations exceeds 90 iterations, while the PPO and MARL algorithms converge after 150 and 170 iterations, respectively. The measured time durations for one iteration of the proposed

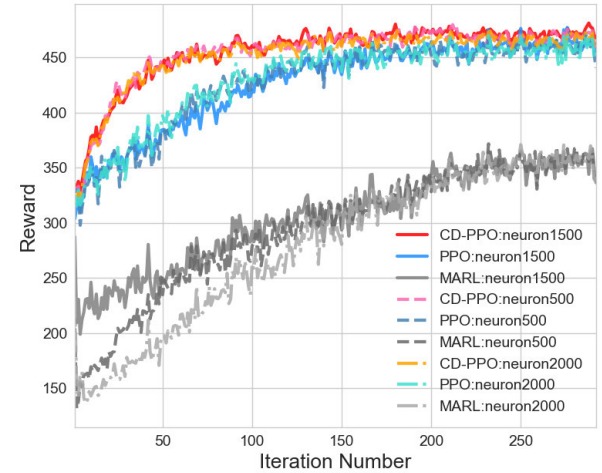


Fig. 7. Algorithm performance using two hidden layers of 1500, 500, and 2000 neurons.

CD-PPO, the PPO and the MARL algorithms are 18.33 seconds, 18.34 seconds, 50.47 seconds on average, respectively, and thus the total amount of training time until convergence are 1,683 seconds, 2,787 seconds, and 8,635 seconds, respectively.

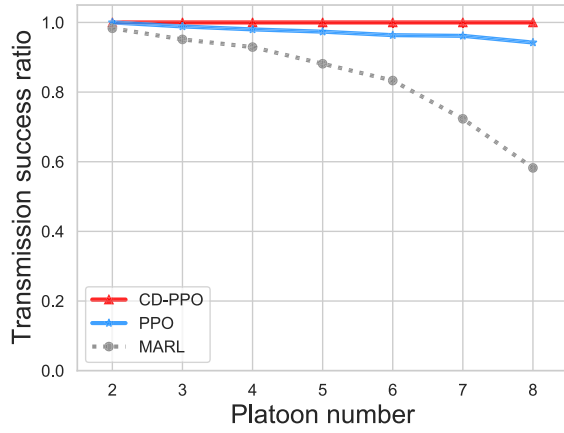


Fig. 8. Transmission success ratio versus the number of platoons for different algorithms.

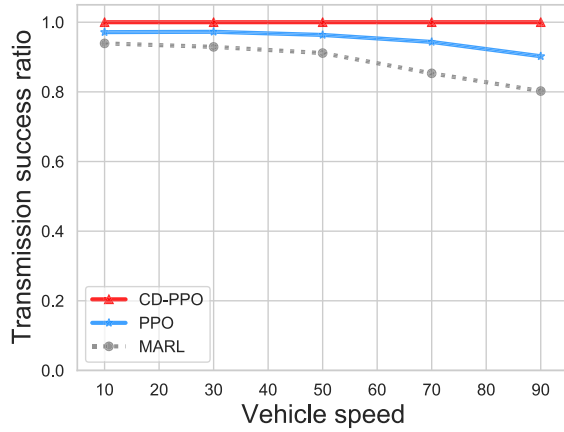


Fig. 9. The relationship between the transmission success ratio of intra-platoon communications and the moving speed of vehicles.

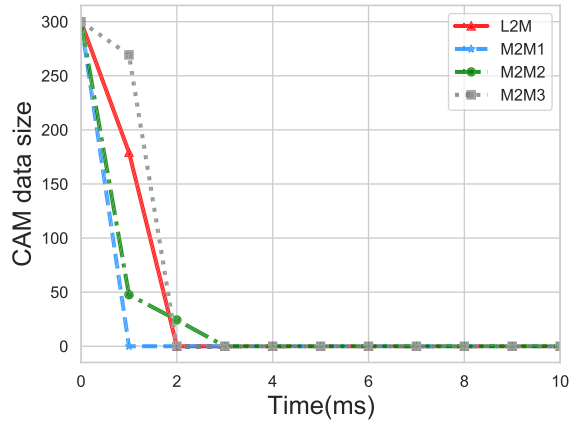


Fig. 10. The remaining CAM data size within $T = 10$ ms for our proposed CD-PPO algorithm in a general vehicular environment.

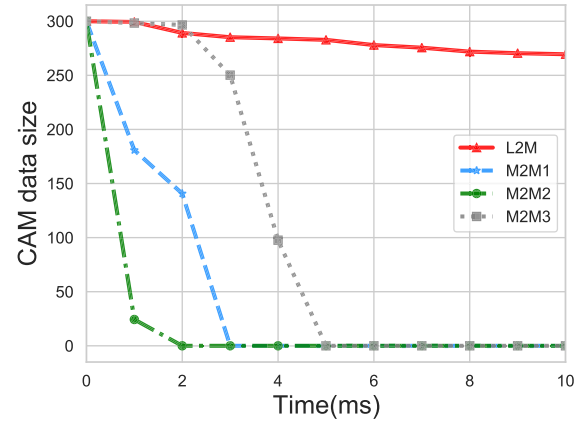


Fig. 11. The remaining CAM data size within $T = 10$ ms for the PPO-based algorithm in a crowded vehicular environment.

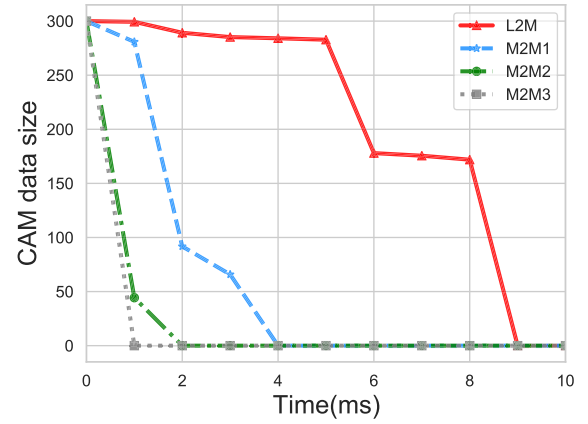


Fig. 12. The remaining CAM data size within $T = 10$ ms for our proposed CD-PPO algorithm in a crowded vehicular environment.

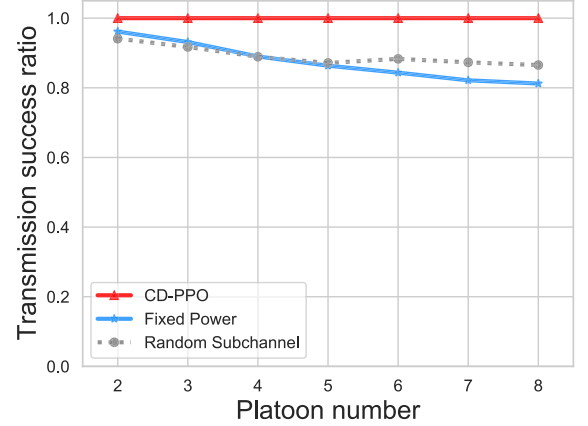


Fig. 13. Transmission success ratio versus the number of platoons for different schemes.

Fig. 6 shows the performance of the proposed CD-PPO algorithm under different learning rates. For comparison, we set the learning rate δ to three values, 0.001 (LR setting 2), 0.0001 (LR setting 1), and 0.00001 (LR setting 3), respectively. It can be observed from Fig. 6 that the proposed CD-PPO algorithm outperforms the other two algorithms in the cases with different learning rates. The proposed algorithm achieves stable performance across different learning rates.

Considering the performance of the MARL algorithm, the learning rate is set to 0.0001.

In order to evaluate the effect of the DNN structure with different neuron numbers on the algorithm performance, we implement two hidden layers with neurons of (1500; 1500), (500; 500), and (2000; 2000), respectively, as shown in Fig. 7. It can be seen that the proposed CD-PPO algorithm and the PPO algorithm have almost the same rewards and convergence

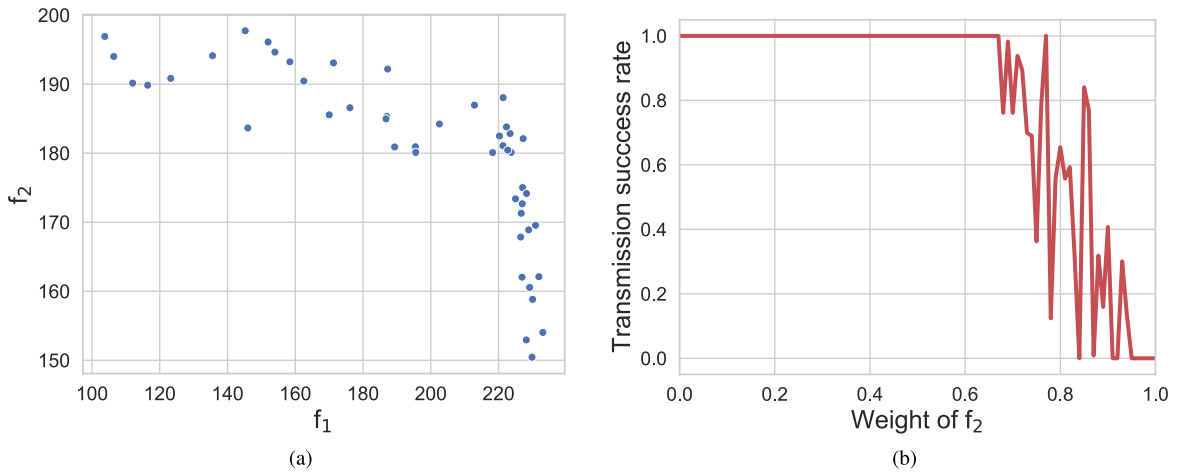


Fig. 14. (a) The approximate Pareto Front using the proposed DRL-based framework; (b) Transmission success ratio versus the weight of the optimization objective f_2 .

speed using different numbers of neurons. For the MARL algorithm, the increase in the number of neurons (e.g., from (500; 500) to (1500; 1500)) leads to more rewards and a faster convergence speed. However, if the number of neurons is too large (e.g., (2000; 2000)), it will lead to a decrease in the reward. The reason is that a large number of neurons may result in the overfitting of the DNN and also raise the computational cost for the neural network training. Therefore, two hidden layers with neurons of (1500; 1500) are chosen in the DNN structure.

We explore the impact of different platoon number on the transmission success ratio of intra-platoon communications. Fig. 8 indicates that the proposed CD-PPO algorithm achieves higher transmission success ratio than two comparison algorithms with varying number of platoons. As the number of platoons increases, the transmission success ratios of the other two algorithms drop, especially for the MARL-based algorithm. When the platoon number equals to 8, the transmission success ratio of MARL-based algorithm is less than sixty percent. In addition, when the number of agents changes, a link-level agent trained with deep Q-learning needs to train a new neural network if the platoon dynamics is considered in advance. Our approach can scale well with the increasing number of platoons. Thus, when platoons increase, it will only enlarge the state-action space and increase the complexity of the action execution to some extent, which can be mitigated by the distributed execution.

In order to evaluate the effect of the vehicle speed on the system performance, transmission success ratio of intra-platoon communications with nine vehicle speeds is shown in Fig. 9. It can be seen that, the transmission success ratio trained with CD-PPO is not affected by the increase of vehicle speed. Additionally, the gap between CD-PPO and the MARL-based algorithm or PPO algorithm becomes larger as vehicle speed increases. This is because the proposed scheme deals with the credit assignment problem using a contribution-based approach.

To verify the performance of the proposed algorithm, we cut out an episode where CD-PPO enables intra-platoon

communications to deliver the CAM data of 300 bytes successfully while the benchmarks fail. Figures (10)-(12) illustrate changes in the remaining CAM data size of all L2M and M2M links within the time constraint (i.e., $T = 10$ ms). Fig. 10 shows the remaining CAM data size for CD-PPO in a general multi-platoon network. It can be seen from Fig. 10 that the data transmission of one L2M link and three M2M links have all been completed in the second time step. Through setting larger rewards α and β after CAM data for L2M and M2M links is delivered completely, we encourage L2M link and M2M link to finish delivery as soon as possible. However, when traffic becomes heavy-crowded as vehicles get closer to each other, serious interference is caused. In this case, the PPO and MARL algorithms cannot complete the data transfer within the requirement of transmission delay. Fig. 11 and Fig. 12 unfold the remaining CAM for the PPO-based algorithm and CD-PPO algorithm in a crowded vehicular environment, respectively. Fig. 11 shows that the 3rd M2M link completes its delivery in the fifth step while the L2M link fails to complete. This is because the failed agent suffering from daunting exploration due to the credit assignment problem. However, as shown in Fig. 12, the proposed CD-PPO can meet the transmission delay requirement in such a crowded vehicular environment. By quantifying the contribution of each agent to the global reward, CD-PPO can alleviate the impact of lazy agents and guarantee system performance.

In Fig. 13, we also compare the transmission success ratio achieved by the following schemes: i) the proposed joint design scheme, ii) a scheme with power allocation optimization only, using a random subchannel selection, and iii) a scheme with subchannel selection optimization only, using a fixed power allocation. It can be seen from Fig. 13 that the proposed joint design scheme outperforms the other two schemes in terms of transmission success ratio of intra-platoon communications. In addition, the transmission success ratio gap between the proposed joint design scheme and the fixed power scheme increases as the number of platoons increases. The reason is that the fixed power scheme results in large interference among V2V links when

the number of platoons increases, thus affecting the system performance.

Fig. 14(a) shows the approximate Pareto Front obtained by the proposed DRL-based framework. Fig. 14(b) illustrates the weight of the second optimization objective f_2 versus the transmission success ratio. When λ_2 is greater than 0.7, although the MOS value of V2I links is high, the transmission success ratio cannot be guaranteed to be 100 percent. A better balance between the two objectives can be achieved when the weighting factor of the second objective is in the range of 0.4 to 0.6. Through these results, we find that the DRL-based framework can handle the multi-objective resource allocation problem effectively and efficiently. Once the trained model is available, it can apply to newly encountered problems without retraining.

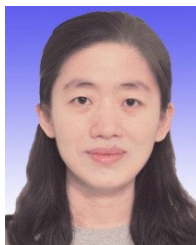
V. CONCLUSION

We investigated the problem of resource allocation in multi-platoon vehicular networks, where V2V communications reuse the spectrum preoccupied by V2I links. Considering the high dynamics in the vehicular environments, we formulated the problem of power allocation and subchannel selection for maximizing the success ratio of intra-platoon transmissions and the MOS of V2I communications as a stochastic game. Besides, we provided a DRL-based framework combined with CD-PPO algorithm to solve the proposed multi-objective optimization problem. Simulation results show that CD-PPO algorithm can improve the reliability of communications in platoons while alleviating interference and strengthening the cooperation among platoons.

REFERENCES

- [1] H. Xu, K. Zhu, J. Ji, and R. Wang, "Deep reinforcement learning for resource allocation in multi-platoon vehicular networks," in *Proc. WASA*, 2021, pp. 1–6.
- [2] F.-Y. Wang, "Parallel control and management for intelligent transportation systems: Concepts, architectures, and applications," *IEEE Trans. Intell. Transp. Syst.*, vol. 11, no. 3, pp. 630–638, Sep. 2010.
- [3] S. Santini, A. Salvi, A. S. Valente, A. Pescape, M. Segata, and R. L. Cigno, "Platooning maneuvers in vehicular networks: A distributed and consensus-based approach," *IEEE Trans. Intell. Vehicles*, vol. 4, no. 1, pp. 59–72, Mar. 2019.
- [4] Y. Li, W. Chen, S. Peeta, and Y. Wang, "Platoon control of connected multi-vehicle systems under V2X communications: Design and experiments," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 5, pp. 1891–1902, May 2020.
- [5] S. Chen et al., "Vehicle-to-everything (V2X) services supported by LTE-based systems and 5G," *IEEE Commun. Standards Mag.*, vol. 1, no. 2, pp. 70–76, Jun. 2017.
- [6] L. Hobert, A. Festag, I. Llatser, L. Altomare, F. Visintainer, and A. Kovacs, "Enhancements of V2X communication in support of cooperative autonomous driving," *IEEE Commun. Mag.*, vol. 53, no. 12, pp. 64–70, Dec. 2015.
- [7] G. Guo and S. Wen, "Communication scheduling and control of a platoon of vehicles in VANETs," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 6, pp. 1551–1563, Jun. 2016.
- [8] R. Molina-Masegosa and J. Gozalvez, "LTE-V for sidelink 5G V2X vehicular communications: A new 5G technology for short-range vehicle-to-everything communications," *IEEE Veh. Technol. Mag.*, vol. 12, no. 4, pp. 30–39, Dec. 2017.
- [9] L. Xu, L. Y. Wang, G. Yin, and H. Zhang, "Communication information structures and contents for enhanced safety of highway vehicle platoons," *IEEE Trans. Veh. Technol.*, vol. 63, no. 9, pp. 4206–4220, Nov. 2014.
- [10] Y. Sun, D. W. K. Ng, Z. Ding, and R. Schober, "Optimal joint power and subcarrier allocation for full-duplex multicarrier non-orthogonal multiple access systems," *IEEE Trans. Commun.*, vol. 65, no. 3, pp. 1077–1091, Mar. 2017.
- [11] W. Sun, E. G. Ström, F. Brännström, K. C. Sou, and Y. Sui, "Radio resource management for D2D-based V2V communication," *IEEE Trans. Veh. Technol.*, vol. 65, no. 8, pp. 6636–6650, Aug. 2016.
- [12] M. I. Ashraf, M. Bennis, C. Perfecto, and W. Saad, "Dynamic proximity-aware resource allocation in vehicle-to-vehicle (V2V) communications," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Dec. 2016, pp. 1–6.
- [13] H. Ye and G. Y. Li, "Deep reinforcement learning for resource allocation in V2V communications," in *Proc. IEEE Int. Conf. Commun.*, May 2018, pp. 1–6.
- [14] M. I. Ashraf, C.-F. Liu, M. Bennis, W. Saad, and C. S. Hong, "Dynamic resource allocation for optimized latency and reliability in vehicular networks," *IEEE Access*, vol. 6, pp. 63843–63858, 2018.
- [15] L. Liang, S. Xie, G. Y. Li, Z. Ding, and X. Yu, "Graph-based resource sharing in vehicular communication," *IEEE Trans. Wireless Commun.*, vol. 17, no. 7, pp. 4579–4592, Jul. 2018.
- [16] S. Wang, T. Lv, W. Ni, N. C. Beaulieu, and Y. J. Guo, "Joint resource management for MC-NOMA: A deep reinforcement learning approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 9, pp. 5672–5688, Sep. 2021.
- [17] S. Wang, C. Yuen, W. Ni, Y. L. Guan, and T. Lv, "Multiagent deep reinforcement learning for cost- and delay-sensitive virtual network function placement and routing," *IEEE Trans. Commun.*, vol. 70, no. 8, pp. 5208–5224, Aug. 2022.
- [18] S. Wang, T. Lv, X. Zhang, Z. Lin, and P. Huang, "Learning-based multi-channel access in 5G and beyond networks with fast time-varying channels," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5203–5218, May 2020.
- [19] H. Ye, L. Liang, G. Y. Li, J. Kim, L. Lu, and M. Wu, "Machine learning for vehicular networks: Recent advances and application examples," *IEEE Veh. Technol. Mag.*, vol. 13, no. 2, pp. 94–101, Jun. 2018.
- [20] L. Liang, H. Ye, and G. Y. Li, "Toward intelligent vehicular networks: A machine learning framework," *IEEE Internet Things J.*, vol. 6, no. 1, pp. 124–135, Feb. 2019.
- [21] Y. He et al., "Deep-reinforcement-learning-based optimization for cache-enabled opportunistic interference alignment wireless networks," *IEEE Trans. Veh. Technol.*, vol. 66, no. 11, pp. 10433–10445, Nov. 2017.
- [22] H. Mao, M. Alizadeh, I. Menache, and S. Kandula, "Resource management with deep reinforcement learning," in *Proc. 15th ACM Workshop Hot Topics Netw.*, Nov. 2016, pp. 50–56.
- [23] M. A. Salahuddin, A. Al-Fuqaha, and M. Guizani, "Reinforcement learning for resource provisioning in the vehicular cloud," *IEEE Wireless Commun.*, vol. 23, no. 4, pp. 128–135, Aug. 2016.
- [24] H. Ye, G. Y. Li, and B.-H. F. Juang, "Deep reinforcement learning based resource allocation for V2V communications," *IEEE Trans. Veh. Technol.*, vol. 68, no. 4, pp. 3163–3173, Apr. 2019.
- [25] L. Liang, H. Ye, and G. Y. Li, "Spectrum sharing in vehicular networks based on multi-agent reinforcement learning," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 10, pp. 2282–2292, Oct. 2019.
- [26] H. Peng et al., "Performance analysis of IEEE 802.11p DCF for multiplatooning communications with autonomous vehicles," *IEEE Trans. Veh. Technol.*, vol. 66, no. 3, pp. 2485–2498, Mar. 2017.
- [27] G. Nardini, A. Virdis, C. Campolo, A. Molinaro, and G. Stea, "Cellular-V2X communications for platooning: Design and evaluation," *Sensors*, vol. 18, no. 5, p. 1527, 2018.
- [28] C. Campolo, A. Molinaro, G. Araniti, and A. O. Berthet, "Better platooning control toward autonomous driving: An LTE device-to-device communications strategy that meets ultralow latency requirements," *IEEE Veh. Tech. Mag.*, vol. 12, no. 1, pp. 30–38, Mar. 2017.
- [29] H. Peng et al., "Resource allocation for cellular-based inter-vehicle communications in autonomous multiplatoons," *IEEE Trans. Veh. Technol.*, vol. 66, no. 12, pp. 11249–11263, Dec. 2017.
- [30] J. Mei, K. Zheng, L. Zhao, L. Lei, and X. Wang, "Joint radio resource allocation and control for vehicle platooning in LTE-V2V network," *IEEE Trans. Veh. Technol.*, vol. 67, no. 12, pp. 12218–12230, Dec. 2018.
- [31] P. Wang, B. Di, H. Zhang, K. Bian, and L. Song, "Platoon cooperation in cellular V2X networks for 5G and beyond," *IEEE Trans. Wireless Commun.*, vol. 18, no. 8, pp. 3919–3932, Aug. 2019.
- [32] *Technical Specification Group Radio Access Networks; Study enhancement 3GPP Support for 5G V2X Service*, document TR 22.886, V15.1.0, Release 15, 3GPP, Mar. 2017.

- [33] M. Rugelj, M. Volk, U. Sedlar, J. Sterle, and A. Kos, "A novel user satisfaction prediction model for future network provisioning," *Telecommun. Syst.*, vol. 56, no. 3, pp. 417–425, Jul. 2014.
- [34] A. Saul and G. Auer, "Multiuser resource allocation maximizing the perceived quality," *EURASIP J. Wireless Commun. Netw.*, vol. 2009, no. 1, pp. 1–15, Dec. 2009.
- [35] L. Busoni, R. Babuska, and S. B. De, "A comprehensive survey of multiagent reinforcement learning," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 38, no. 2, pp. 156–172, Feb. 2008.
- [36] Y. S. Nasir and D. Guo, "Multi-agent deep reinforcement learning for dynamic power allocation in wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 10, pp. 2239–2250, Oct. 2019.
- [37] J. Foerster et al., "Stabilising experience replay for deep multi-agent reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1146–1155.
- [38] H. Meshgi, D. Zhao, and R. Zheng, "Optimal resource allocation in multicast device-to-device communications underlaying LTE networks," *IEEE Trans. Veh. Technol.*, vol. 66, no. 9, pp. 8357–8371, Sep. 2017.
- [39] M. Rugelj, U. Sedlar, M. Volk, J. Sterle, M. Hajdinjak, and A. Kos, "Novel cross-layer QoE-aware radio resource allocation algorithms in multiuser OFDMA systems," *IEEE Trans. Commun.*, vol. 62, no. 9, pp. 3196–3208, Sep. 2014.
- [40] K. Li, T. Zhang, and R. Wang, "Deep reinforcement learning for multiobjective optimization," *IEEE Trans. Cybern.*, vol. 51, no. 6, pp. 3103–3114, Jun. 2021.
- [41] M. Tan, "Multi-agent reinforcement learning: Independent vs. cooperative agents," in *Proc. 10th Int. Conf. Mach. Learn.*, 1993, pp. 330–337.
- [42] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, "Trust region policy optimization," in *Proc. 32nd Int. Conf. Mach. Learn.*, 2015, pp. 1889–1897.
- [43] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
- [44] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," 2015, *arXiv:1506.02438*.
- [45] D. Ye et al., "Mastering complex control in MOBA games with deep reinforcement learning," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 4, pp. 6672–6679.
- [46] *Enhanced LTE Support for Aerial Vehicle*, document 36.777, Release 15, Dec. 2017.
- [47] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8026–8037.



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